The Roles of Policy, Labor, and Perception of the Risk of Using Opioids on Demand for Opioids

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Motivation

- Rules on opioid prescribing and Must-Access Prescription Drug Monitoring Programs have been introduced recently.
  - The purpose is to decrease opioid use disorder and overdose death.
  - The motivation is that excessive opioid prescribing caused the opioid crisis, and people have a bad idea about how risky "misusing" opioids is.
  - It is well-documented that such programs have increased overdose deaths.
  - Yet, opioid "misuse" has decreased overall by these policies.
- The policy's heterogeneous effect across the population on opioid "misuse," Opioid Use Disorder, and death is understudied.
- This paper focuses on the interaction of opioid "misuse" and labor as a channel through which the policy's effect propagates.
- This paper also highlights the role of the perception of the risk of misusing opioids as amplifying the policy's effect on opioid "misuse".

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Research Question

- How does the stringency in restrictions on prescribing opioids affect opioid "misuse", opioid use disorder, and death?
- What counterfactual policies can decrease opioid "misuse", opioid use disorder, and death altogether?

Aim of Paper

- Provide novel associations across:
  - Opioid misuse, job displacement, and state-level policies
  - Perception of the risk of misusing opioids, job displacement, and state-level policies
- Develop a dynamic discrete choice model of opioid use and work
  - To reveal heterogeneous substitution toward illegal opioids affected by policies.
  - To measure the magnitude of the perception bias in the risk of misusing opioids.
- Predict counterfactual policies' effect on opioid "misuse," Opioid Use Disorder, and death
  - Illegal opioids market: increase prices for illegally traded opioids
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Contribution

- This paper evaluates the heterogeneous effect of health policies via labor.
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  - Health policies: Kim [2021], Schnell [2022], Balestra et al. [2023], Sacks et al. [2021], Beheshti [2022], Mallatt [2022], Alpert et al. [2018]
- This paper provides a novel framework of dynamic demand on experience goods under conditionally homogeneous beliefs.
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Definitions

In the past 12 months:

- Nonuser: did not use opioids
- Prescription user: has been prescribed opioids and used them as directed by a doctor
- Opioid "misuser": used opioids not as directed by a doctor in any way
  - with his prescription
  - without his prescription, i.e., obtained in the illegal market
- Opioid Use Disorder (OUD): suffers from opioid use, including adverse symptoms, doing routine tasks at home, work, etc. (DSM-IV)

Policy Background

- State-level policies on prescribing opioids
  - Maximum number of days, amount, etc.
  - Aggregated to represent the stringency for all prescriptions on adults
- Must-Access Prescription Drug Monitoring Program
  - Requires doctors to check patient's prescription history
  - Decreases prescription on the extensive margin (e.g., Sacks et al. [2021])
- In this paper, policies affect prescription rates on the extensive margin.

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Data

- Opioid use and labor
  - **NSDUH**: Restricted National Survey of Drug Use and Health, 2015-2019
  - **NVSS**: Restricted National Vital Statistics System-Multiple Causes of Death, 2000-2019
  - **MEPS**: Public Medical Expenditure Panel Survey, 2015-2019
  - **SIPP**: Survey of Income and Program Participation, 2015-2019
- Opioid prescribing policy and opioid prices
  - **PDAPS**: Prescription Drug Abuse Policy System
  - **StreetRx**: 2013-2019

Data Patterns

- Opioid "misuse" is:
  - Positively associated with displacement from labor and bad health.
  - Negatively associated with perceiving a higher risk of "misusing" opioids.
  - Prevalent in working age, while prescription is prevalent among the elderly.

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Dynamic Model of Work and Opioid Use

- In each period, the individual observes his state's policies on prescribing opioids and opioid prices.
- He observes his work experience, displacement from labor, health, and opioid use disorder. He is also prescribed opioids based on prescription policies.
- The perception of the risk of "misusing" opioids is formed stochastically.
- He chooses work and opioid "misuse," but:
  - He cannot work during this period if displaced from labor.
  - He can only "misuse" illegal opioids if he is not prescribed opioids.
- He may experience opioid use disorder based on his exposure to opioids.
- He faces a risk of death in each period; if he survives, he ages a year.
  - After age 60, he retires if he does not work or is displaced from labor. He expects that he will not return to work.
  - At age 65, he stops aging and solves an infinite-horizon problem until death.

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  - After age 60, he retires if he does not work or is displaced from labor. He expects that he will not return to work.
  - At age 65, he stops aging and solves an infinite-horizon problem until death.

State Space

Dynamic Model of Work and Opioid Use

- In each period, the individual observes his state's policies on prescribing opioids and opioid prices.
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- A representative individual living in state g with education e
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Figure: State Realization and Decision in Each Period

$$\Omega_g = (g, r_{g,t}, p_t^{rx}, p_{g,t}^{il})$$

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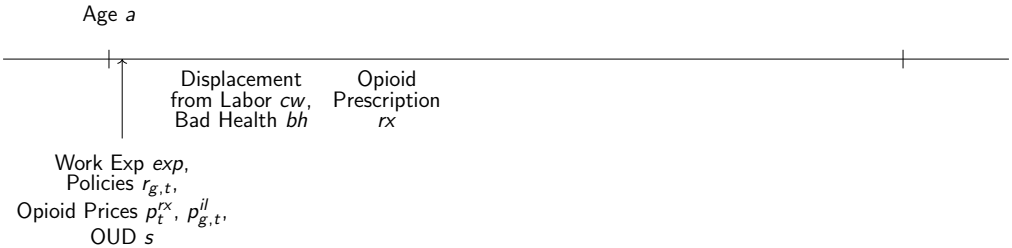

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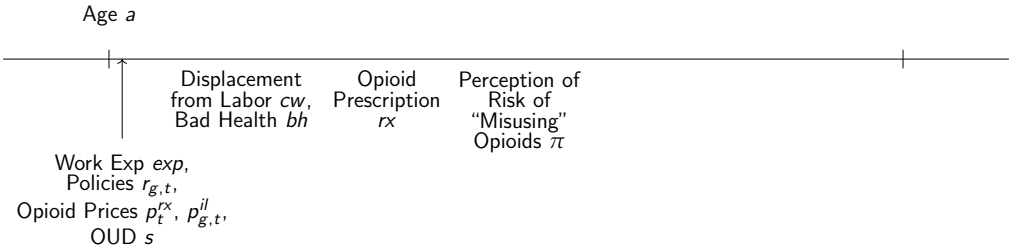

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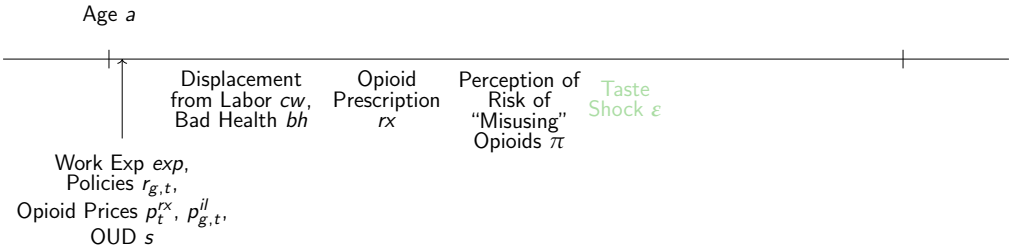

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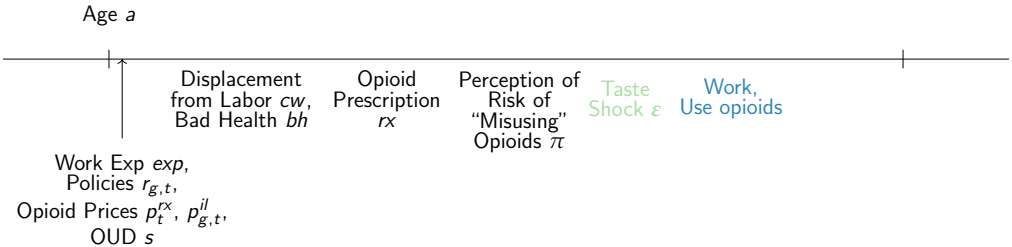

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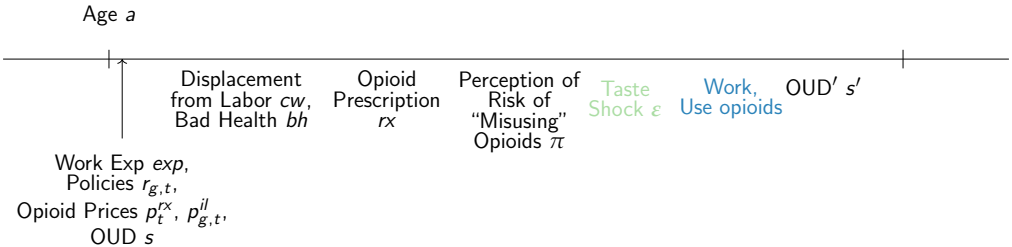

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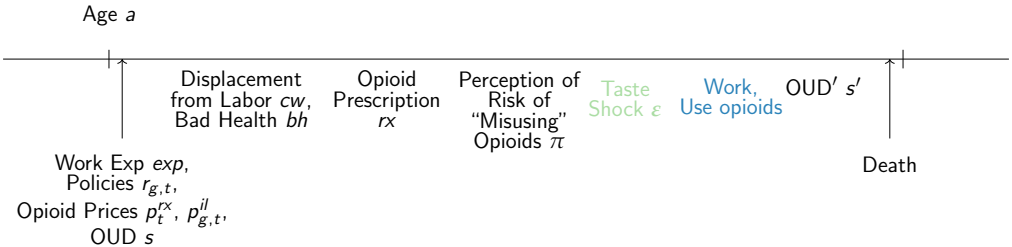

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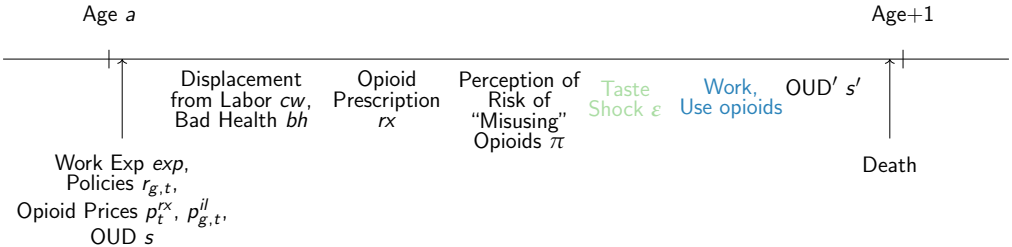

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Choice Set

- The individual can work and "misuse" opioids each period:

$$d_w = \mathbf{1}\{\text{work}\}$$
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- The choice set is affected by i) displacement from labor and ii) being prescribed opioids.
  - If he is displaced from labor, then he cannot work in this period.
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Formal Table

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Dynamic Model of Work and Opioid Use

Per-period Utility

$$u_j(\Omega_a, s, \pi, \varepsilon; \boldsymbol{\theta}^y, \boldsymbol{\theta}^w, \boldsymbol{\theta}^o) =$$
$$\underbrace{\bar{y}(\Omega_a; \boldsymbol{\theta}^y)}_{\text{Baseline Income}} + \underbrace{y(\Omega_a, s, d_o^{rx}, d_o^{il}; \boldsymbol{\theta}^w) d_w}_{\text{Income from Working}} + \underbrace{u_o(\Omega_a, \Omega_g, s, d_w, d_o^{rx}, d_o^{il}; \boldsymbol{\theta}^o)}_{\text{Utility from "misusing" opioids}} + \varepsilon_j.$$

- Baseline income approximates lifetime earnings.
- Income from work reflects productivity loss from OUD and opioid "misuse."
- Utility from "misusing" opioids captures:
  - Compensating for the disutility of being displaced from labor
  - The disutility of not "misusing" opioids if one has Opioid Use Disorder
  - The complementarity of "misusing" opioids and not working

Dynamic Model of Work and Opioid Use

Transition Probabilities

- Suffering from OUD s'
- Death
- Displacement from work & bad health (cw', bh')
- Being prescribed opioids rx'
- Perception of the risk of misusing opioids π'

$$f_s(s, rx, d_o^{rx}, d_o^{il}; \boldsymbol{\theta}^s) = \begin{cases} 0 & \text{if } rx = d_o^{rx} = d_o^{il} = 0 \\ \frac{\exp(x_s \boldsymbol{\theta}^s)}{1+\exp(x_s \boldsymbol{\theta}^s)} & \text{o.w.} \end{cases}$$

where $x_s = [s, rx, d_o^{rx}, d_o^{il}, d_o^{rx} d_o^{il}]$

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$$f_d(s', a, e, d_{rx}, d_{il}; \theta^d) = \sum_{i \in D} f(dth_i | s', a, e, g, d_{rx}, d_{il}; \boldsymbol{\theta}_i^d)$$

$$D = \{\text{OCD}, \text{OD:Rx}, \text{OD:Illegal}, \text{OD:Rx \& Illegal}\}$$

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$$f_{ch}((cw', bh') = i | x_{ch}; \boldsymbol{\theta}^{ch})$$

$$= \frac{\exp(x_{ch}\theta_i^{ch} + \alpha_g)}{1 + \sum_{i=1}^{7} \exp(x_{ch}\theta_i^{ch} + \alpha_g^{ch})}, i \neq (0,0)$$

$$x_{ch} = [\mathbf{1}_{cw=1}, \mathbf{1}_{cw=2}, \mathbf{1}_{cw=3}, bh, rx, a, e, exp, d_w, s']$$

α_g^{ch} state-FE

Dynamic Model of Work and Opioid Use

Transition Probabilities

- Suffering from OUD s'
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$$f_{rx}(x_{rx}, \alpha_g, r_g; \boldsymbol{\theta}^{rx}) = \frac{\exp(x_{rx}\theta^{rx} + \gamma r_g + \gamma_g)}{1 + \exp(x_{rx}\theta^{rx} + \gamma r_g + \alpha_g^{rx})}$$

$$x_{rx} = [\mathbf{1}_{cw'=1}, \mathbf{1}_{cw'=2}, \mathbf{1}_{cw'=3}, bh', rx, a, e, s']$$

α_g^{rx} state-FE

Dynamic Model of Work and Opioid Use

Transition Probabilities

- Suffering from OUD s'
- Death
- Displacement from work & bad health (cw', bh')
- Being prescribed opioids rx'
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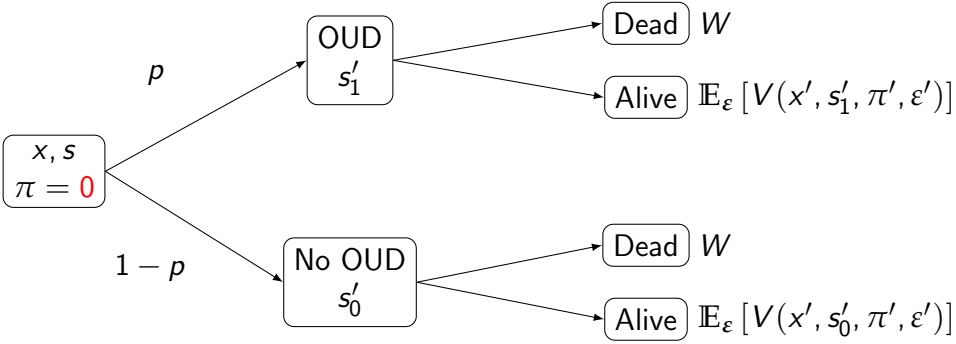
$$f_\pi(x_\pi; \boldsymbol{\theta}^\pi) = \frac{\exp(x_\pi \theta^\pi)}{1 + \exp(x_\pi \theta^\pi)}$$

$$x_\pi = [\mathbf{1}_{cw'=1}, \mathbf{1}_{cw'=2}, \mathbf{1}_{cw'=3}, bh', rx', s']$$

Dynamic Model of Work and Opioid Use

Perception of the risk of "misusing" opioids

- The individual stochastically perceives the risk of "misusing" opioids, π.
  - High risk ($\pi = 0$): expects the actual probability of getting OUD
  - Low risk ($\pi = 1$): expects a lower probability of getting OUD by δ



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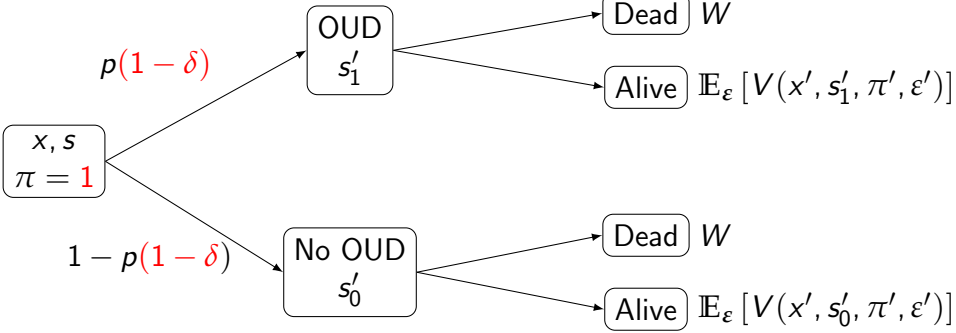

Detail

Dynamic Model of Work and Opioid Use

Value Function Representation

- The value function of a living individual $V(x, s, \pi, \varepsilon)$ is

$$V(x, s, \pi, \varepsilon) = \max_{j \in \mathcal{J}(rx,k)} u_j(x, s; \boldsymbol{\theta}^y, \boldsymbol{\theta}^o) + \varepsilon_j +$$

$$\beta \left[\begin{array}{r} (1-\pi\delta) f_{s_1'|j} \left(f_{d|s_1',j} W + (1 - f_{d|s_1',j}) \sum_{x',\pi'} \mathbb{E}_\varepsilon \left[V(x', s_1', \pi', \varepsilon') \right] f_{x'|x,s_1',d_w} \right) \\ +[1-(1-\pi\delta) f_{s_1'|j})] \left(f_{d|s_0',j} W + (1 - f_{d|s_0',j}) \sum_{x',\pi'} \mathbb{E}_\varepsilon \left[V(x', s_0', \pi', \varepsilon') \right] f_{x'|x,s_0',d_w} \right) \end{array} \right]$$

- W: value of (dis)utility upon death
- δ: magnitude of discounting the probability of getting opioid use disorder
- The model is solved by contraction mapping and backward induction.

Alternative

Identification

The parameters to identify are:

- Initial type probability $\bar{q}(s|x)$
- Per-period utility
  - Income process $\boldsymbol{\theta}^y$, σ_y
  - Utility from "misusing" opioids $\boldsymbol{\theta}^o$
- Transition probabilities
  - Opioid Use Disorder $\boldsymbol{\theta}^s$
  - Death $\boldsymbol{\theta}^d$
  - Displacement from work and bad health $\boldsymbol{\theta}^{ch}$, α_g^{ch}
  - Prescription to opioids $\boldsymbol{\theta}^{rx}$, γ, α_g^{rx}
  - Perception of the risk of misusing opioids $\boldsymbol{\theta}^{\pi}$
- (Dis)utility of death W
- Perception bias δ
- Discount factor $\beta = 0.96$
- Distribution of $\boldsymbol{\varepsilon}$: T1EV

Identification

First stage

If $\bar{q}(s|x)$ is known, the parameters and CCPs below are identifiable:

- Perception of the risk of misusing opioids $\boldsymbol{\theta}^{\pi}$ by concavity of logit
- Unrestricted CCPs $P_j(x, s, \pi)$ under rank condition
- The income process $\boldsymbol{\theta}^{y}$, σ_y by concavity of likelihood
- Transition to Opioid Use Disorder $\boldsymbol{\theta}^{s}$ by concavity of logit

Then, the transition parameters below are identifiable by concavity of logit

- Death $\boldsymbol{\theta}^{d}$
- Displacement from work and bad health $\boldsymbol{\theta}^{ch}$, α_g^{ch}
- Prescription to opioids $\boldsymbol{\theta}^{rx}$, γ, α_g^{rx}

Challenge

Identification

Challenge

- s is not observed; use three proxies to identify $\bar{q}(s|x)$
  - $ev \in \{0, 1\}$: "Ever misused opioids in the past"
  - $\pi_1^* \in \{0, 1\}$: "great risk" of using marijuana
  - $\pi_2^* \in \{0, 1\}$: "great risk" of using cocaine
- NSDUH '15-'19 missing s' if $d_o^{rx} = d_o^{il} = 0$:
  - Use NSDUH '21 for $f_s(s'|rx = 1, d_o^{rx} = 0, d_o^{il} = 0)$
- Assumption:
  - Proxy variables ev, π_1^*, and π_2^* have logistic probability conditional on (x, s)
  - Conditional on (x, s), ev, π_1^*, and π_2^* are independent
  - $f_{ev}(ev = 1|s = 1, x) > f_{ev}(ev = 1|s = 0, x)$ for all x
- Then, $\bar{q}(s|x)$, f_{ev}, f_{π_1}, f_{π_2} are identified. (Hwang [2021])

Transition

Identification

- NSDUH is an RCS: need auxiliary panel data for $\boldsymbol{\theta}^d$, $\boldsymbol{\theta}^{ch}$, $\boldsymbol{\theta}^{rx}$, γ, α_g^{ch}, α_g^{rx}
- NSDUH + NVSS, MEPS, and SIPP
  - NVSS: Death $\boldsymbol{\theta}^d$
  - SIPP: Constraint to work & bad health $\boldsymbol{\theta}^{ch}$
  - MEPS: Opioid Prescription $\boldsymbol{\theta}^{rx}$
  - NSDUH 2015-2019: Year-by-year distribution

Details Limited Feedback

Identification

Hotz-Miller Inversion and Finite Dependence

(Dis)utility of death W, Utility from "misusing" opioids $\boldsymbol{\theta}^o$, Perception bias δ

- Hotz-Miller inversion + finite dependence (Arcidiacono and Miller [2019])
- $\mathbb{E}_{\varepsilon}(V(x,s,\pi,\boldsymbol{\varepsilon})) = v_j(x,s,\pi) + \psi_j(\mathbf{p}(x,s,\pi))$ where $v_j(x,s,\pi)$ is choice-specific conditional value function (CVF)

$$v_j(x,s,\pi;\boldsymbol{\theta}^o, W, \delta) := u_j(x,s;\boldsymbol{\theta}^y,\boldsymbol{\theta}^o) + \beta \left[\begin{array}{r} (1-\pi\delta) f_{s_1'|j} \left(f_{d|s_1',j} W + (1-f_{d|s_1',j}) \sum_{x',\pi'} \bar{V}(x',s_1',\pi') f_{x'|x,s_1',d_w} \right) \\ +[1-(1-\pi\delta) f_{s_1'|j})] \left(f_{d|s_0',j} W + (1-f_{d|s_0',j}) \sum_{x',\pi'} \bar{V}(x',s_0',\pi') f_{x'|x,s_0',d_w} \right) \end{array} \right]$$

$$\psi_j(\mathbf{p}(x,s,\pi)) = \gamma - \log P_j(x,s,\pi) \text{ if } \boldsymbol{\varepsilon} \sim \text{T1EV}$$

Identification

Hotz-Miller Inversion and Finite Dependence

- Mapping the CCPs to CVF differences (Hotz and Miller [1993]): CVF-ω

$$\log \frac{P_j(x, s, \pi)}{P_1(x, s, \pi)} = v_j(x, s, \pi; \boldsymbol{\theta}^o, W, \delta) - v_1(x, s, \pi; \boldsymbol{\theta}^o, W, \delta) \tag{1}$$

- If transition matrices exhibit finite dependence, we cancel out the future ex-ante value functions. Only $\boldsymbol{\theta}^o$, W, δ are unknown. HM-Inversion-FD
- Challenge: δ is unknown and affects finite dependence path.

Identification

- (New!) If there exists a sub-finite dependence path for a subset of state transitions where the perception stayed "great risk" for two periods $(\pi = 0, \pi' = 0)$, then a finite dependence path exists as a function of the weights for the sub-finite dependence path, δ, and $f(s_1|x, s, \pi_0)$.
- $\boldsymbol{\theta}^o, W, \delta$ identified from a set of (1) with finite dependence, given all other parameters for transition probabilities and CCPs HM-Inversion-FD
- The variations for identification are:
  - $\boldsymbol{\theta}^o$: CVF differences at (x, s) across choices
  - W: CVF differences between d_o and $\tilde{d}_o$ across ages
  - δ: CVF differences between d_o and $\tilde{d}_o$ across different π's

Estimation Strategy

A 'two-step' EM algorithm in Arcidiacono and Miller [2011] using finite dependence Arcidiacono and Miller [2019]:

- Stage 1: Estimate the unrestricted CCPs and the transition probabilities
  - EM algorithm for θ^u, $P_j(x, s, \pi)$, θ^π, θ^y, θ^s Detail
  - GMM for θ^d, θ^{ch}, θ^{rx} conditional on $P_j(x, s, \pi)$, θ^u, θ^y, θ^s Detail
- Stage 2: Estimate θ^o, W, δ
  - Guess δ.
  - Find a finite dependence path for each state s.t. (π_0, π_0') Detail
  - Find finite dependence paths for each state with other (π, π') with δ
  - Construct a sufficient set of moments to identify θ^o and W Detail
  - Estimate θ^o, W, δ with MDE Detail
  - Update δ and repeat until convergence.

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Conclusion

- Policy evaluation for restrictions on opioid prescribing
  - Focus on opioid "misuse" and its interaction with labor
  - Redistribution effect on opioid misuse, creating losers to the policy
- A dynamic model to quantify the substitution in opioid use and predict the individual's responses to counterfactual policies
  - Stricter policies? Higher illegal (or prescription) opioid prices?
  - Unemployment benefits to people with opioid use disorder?
  - Raise the awareness of the risk of misusing opioids?

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Conclusion

"When you add hopelessness, unemployment, and lack of education on top of all that, it's kind of a recipe for disaster."

- Jan Rader, Huntington Fire Department Chief, *Heroin(e)*

Disclaimer

- Census: Any views expressed are those of the authors and not those of the U.S. Census Bureau. The Census Bureau has reviewed this data product to ensure appropriate access, use, and disclosure avoidance protection of the confidential source data used to produce this product. This research was performed at a Federal Statistical Research Data Center under FSRDC Project Number 2124.
- NCHS: NCHS approval reflects the judgment that this research, as described in the amended proposal, is a permissible use of the requested data and that it is feasible to complete the project using the RDC. Approval does not mean that NCHS endorses the merit of the proposed research or its substantive, methodological, theoretical, or policy relevance. Approval of a proposal does not explicitly or implicitly guarantee that all output generated by the analysis will be released. Output that poses a disclosure risk will be suppressed.

Opioid Rx, Misuse, OUD, and Overdose

	12-17	18-25	26-34	35-49	50-64	65+
Nonuser	81.56	70.76	66.77	65.47	62.51	64.07
Rx User	15.29	22.39	26.62	29.95	34.17	34.55
Misuser:						
- no OUD	2.71	5.80	5.24	3.65	2.65	1.20
- OUD	0.44	1.05	1.37	0.93	0.67	0.18
Total	100.00	100.00	100.00	100.00	100.00	100.00
Death						
Opioid Overdose	0.0005	0.0125	0.0262	0.0241	0.0180	0.0034
Other Causes of Death	0.0256	0.0733	0.1005	0.2054	0.7244	4.0101
Total	0.0261	0.0858	0.1267	0.2294	0.7424	4.0135

Table: Percentages of Nonuser, Rx User, Misuser, OUD, and Overdose deaths, NSDUH and NVSS: 2015-2019

Back By Year

Opioid Rx, Misuse, OUD, and Overdose

	2015	2016	2017	2018	2019
Nonuser	61.88	64.36	65.07	66.74	68.73
Rx User	33.29	31.11	30.70	29.37	27.44
Misuser:					
- no OUD	3.90	3.68	3.41	3.13	3.18
- OUD	0.93	0.85	0.81	0.77	0.64
Total	100.00	100.00	100.00	100.00	100.00
Death					
Opioid Overdose	0.0131	0.0166	0.0186	0.0181	0.0191
Other Causes of Death	1.0506	1.0492	1.0653	1.0674	1.0643
Total	1.0637	1.0658	1.0839	1.0855	1.0835

Table: Percents of Nonuser, Rx User, Misuser, and OUD. NSDUH & NVSS: 2015-2019

Back By Age

Opioid Death Rates across Age

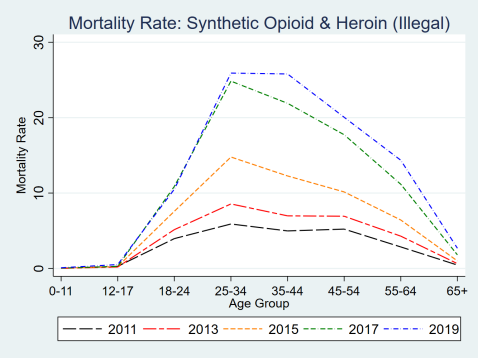

Figure: Mortality rates by illegal opioids across age groups, NVSS, 2011-2019

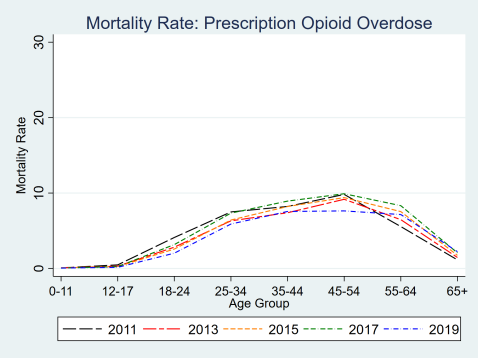

Figure: Mortality rates by prescription opioids across age groups, NVSS, 2011-2019

Back

Opioid Use and Constraints to Work

- Constraint to work is associated with opioid use
  - "laid off": higher misuse of opioids
  - "unable to work": higher Rx

	No Distruption	Unable to Work	Laid Off	All
Nonuser	67.00	35.87	64.96	65.38
Rx user	29.12	57.41	26.52	30.36
Misuser:				
- no OUD	3.26	4.55	5.93	3.46
- OUD	0.62	2.17	2.59	0.80
Total	100.00	100.00	100.00	100.00

Table: Percentages of Nonuser, Rx User, Misuser, and OUD by constraints to work, NSDUH PUF: 2015-2019

Row Back

Opioid Use and Constraints to Work

	No Distruption	Unable to Work	Laid Off	All
Nonuser	92.21	2.66	5.13	100.00
Rx User	86.32	9.17	4.51	100.00
Misuse:				
- no OUD	84.76	6.38	8.86	100.00
- OUD	70.11	13.17	16.72	100.00
Total	89.99	4.85	5.16	100.00

Table: Percentages of Nonuser, Rx User, Misuser, and OUD by constraints to work, NSDUH PUF: 2015-2019

Col Back

Data Patterns

Fact 1: Opioid "Misuse" is associated with Displacement from Work

	Illegal Opioid	
Laid Off	0.675	(0.077)
Unable	0.299	(0.131)
Retired	-0.223	(0.332)
No Great Risk	0.466	(0.058)
Bad Health	0.519	(0.080)

N=82k. Controls: Age category. Standard errors in parentheses.
The sample is male adults who are **not** prescribed opioids.

Table: Logit Regression of Illegal Opioid Use, NSDUH, 2015-2019.

- People who are laid off, in bad health, and who perceive a lower risk of "misusing" opioids are more likely to "misuse" opioids.

Back mlogit

Data Patterns

Fact 1: Opioid "Misuse" is associated with Displacement from Work

	Rx Misuse		Rx & Illegal	
Laid Off	0.365	(0.124)	0.614	(0.147)
Unable	-0.011	(0.153)	-0.102	(0.219)
Retired	-0.528	(0.294)	0.754	(1.105)
Bad Health	0.366	(0.105)	0.429	(0.147)
College	-0.066	(0.094)	-0.451	(0.133)
Opi: No Great Risk	0.361	(0.095)	0.883	(0.106)

N=23k. Controls: Age category. Standard errors in parentheses.
The sample is male adults who are prescribed opioids.

Table: Multinomial Logit Regression of Prescribed Opioid "Misuse" and Misusing both Prescribed and Illegal Opioids, NSDUH, 2015-2019.

- People who are laid off, in bad health, and who perceive a lower risk of "misusing" opioids are more likely to "misuse" opioids.

Back logit

Data Patterns

Fact 2: The Perception of Risk of Misusing Opioids is Associated with Displacement from Labor

	Opioids: Not a Great Risk	
Laid Off	0.247	(0.059)
Unable	0.169	(0.083)
Retired	0.354	(0.060)
Prescribed Opioids	-0.213	(0.035)
Bad Health	0.10	(0.051)
Cocaine: Not a Great Risk	3.087	(0.045)
Marijuana: Not a Great Risk	0.781	(0.083)
Ever Misused	0.192	(0.169)

N=115k. Controls: Age Category. Standard errors in parentheses. The sample is male adults.

Table: Logit Regression of Not Perceiving a Great Risk of Misusing Opioids, NSDUH, 2015-2019.

- People who are laid off, unable to work, and in bad health tend to perceive a lower risk of "misusing" opioids.

Data Patterns

Fact 3: Opioid "Misuse" and overdose death are prevalent in working age group

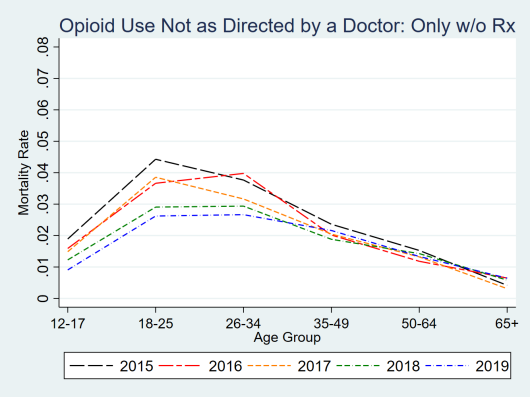

Figure: Opioid "Misuse": Illegal Opioids Only

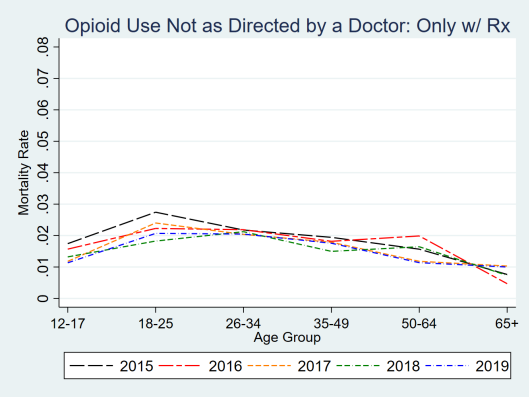

Figure: Opioid "Misuse": Prescription Opioids Only

Back ▸ OD

Data Patterns

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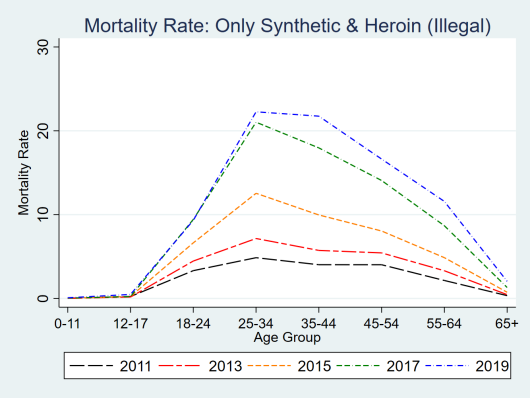

Figure: Opioid Overdose deaths: Illegal Opioids Only

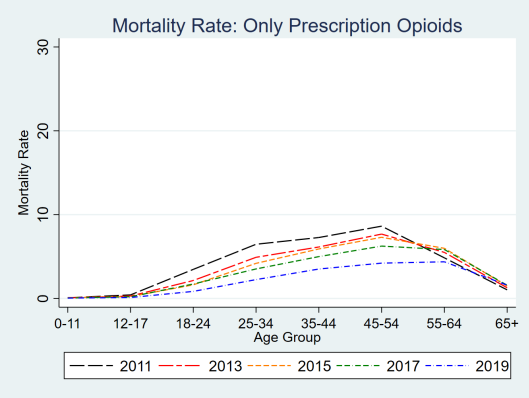

Figure: Opioid Overdose deaths: Prescription Opioids Only

Back ▸ Misuse

State-level Restrictions and Rx Opioids

- I use aggregate data to proxy for opioid Rx rate ($y_{i,t}$):
  - Rx/100: Opioid Rx per 100 population in state g in year t
  - MME[1]: The amount of opioids dispensed in state g in year-**quarter** t
- Two-way fixed effects (TWFE) regression with 2000-2019 data:

$$\log(y_{g,t}) = \beta_0 r_{g,t} + \sum_{k=1}^{3} \beta_j f\{cw = k\}_{g,t} + \beta_4 f\{bh\}_{g,t} + \alpha_g + \delta_t + \varepsilon_{g,t}$$

- The coefficient β are negative for both regressions.

Back

State-level Restrictions and Illegal Opioids

Prices for illegally traded opioids: 2014-2019

- Are street prices correlated with the state-level restrictions on prescription opioids?
- Average of the reported price per MME in state g and year t during 2013-2019

$$\bar{p}_{g,t} = \beta_0 r_{g,t} + \sum_{k=1}^{3} \beta_j f\{cw = k\}_{g,t} + \beta_4 f\{bh\}_{g,t} + \alpha_g + \delta_t + \varepsilon_{g,t}$$

- The coefficient β is statistically insignificant.

Back

State-level Restrictions and Mortality Rates

Mortality rates by types of opioid

- Mortality Rate = Number of Deaths / Population * 100,000
- Two-way fixed effect regression:

$$\text{Mortality Rate}_{g,t} = \beta_0 r_{g,t} + \sum_{k=1}^{3} \beta_j f\{cw = k\}_{g,t} + \beta_4 f\{bh\}_{g,t} + \alpha_g + \delta_t + \varepsilon_{g,t}$$

- The coefficients β are positive for synthetic opioids, negative for heroin, and (insignificant) positive for Rx opioids.

Back

Choice Set

- An individual can choose to work or misuse opioids each period:

$$d_w = \mathbf{1}\{\text{work}\}$$
$$d_o^{rx} = \mathbf{1}\{\text{misuse Rx opioids}\}$$
$$d_o^{il} = \mathbf{1}\{\text{use illegal opioids}\}$$

- An action is indexed by $j = 1 + d_o^{rx} + 2d_o^{il} + 4d_w$ and $d_j = \mathbf{1}\{\text{Chooses } j\}$.
- The choice set $\mathcal{J}(rx, k)$ is defined by i) being prescribed opioids rx and ii) displacement from labor market $k = \mathbf{1}\{cw \neq 0\}$:

$$\mathcal{J}(0,0) = \{1,3,5,7\}$$
$$\mathcal{J}(1,0) = \{1,2,4,5,6,8\}$$
$$\mathcal{J}(0,1) = \{1,3\}$$
$$\mathcal{J}(1,1) = \{1,2,4\}$$

Back Table State Space

Choice Set

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$$d_w = \mathbf{1}\{\text{work}\}$$
$$d_o^{rx} = \mathbf{1}\{\text{misuse Rx opioids}\}$$
$$d_o^{il} = \mathbf{1}\{\text{use illegal opioids}\}$$

- Denote $\tilde{d}_j$ as a negation of action d_j.

(cw, rx)	d_w	$d_o^{rx}, \tilde{d}_o^{il}$	$\tilde{d}_o^{rx}, d_o^{il}$	d_o^{rx}, d_o^{il}
(0,0)	✓	✗	✓	✗
(0,1)	✓	✓	✗	✓
(1,0)	✗	✗	✓	✗
(1,1)	✗	✓	✗	✓

Table: Choice Set

Back Formal State Space

Dynamic Model of Work and Opioid Use

State Variables

- Age $a = 23, \ldots, 65$, state g
- State-level restrictions $r_g \in \{0, 1\}$, Rx opioid price p^{rx}, illegal opioid price p_g^{il}
- Work experience $exp \in \{0, 1\}$, education $e \in \{0, 1\}$
- Bad health $bh \in \{0, 1\}$
- Displacement from work: $cw \in \{0, 1, 2, 3\}$
  - 1: laid off, 2: ill/disabled, 3: retired.
  - $k := \mathbf{1}\{cw \neq 0\}$
- Prescribed opioids $rx \in \{0, 1\}$, OUD $s \in \{0, 1\}$
- Perception of the risk of "misusing" opioids $\pi \in \{0, 1\}$
- Idiosyncratic shocks to preference $\boldsymbol{\varepsilon} := (\varepsilon_1, \ldots, \varepsilon_{J(rx,k)})$

Choice Set

Derivation: CCP

$$P(d_j|x,\pi,\pi^*,ev) = \sum_s q(s|x,\pi,\pi^*,ev)P(d_j|x,\pi,s)$$

$$= \sum_s \frac{P(\pi|x,s,\pi^*,ev;\boldsymbol{\theta}^\pi)q(s|x,\pi^*,ev)}{P(\pi|x,\pi^*,ev)}P(d_j|x,\pi,s) \quad \because \text{Bayes' Rule}$$

$$= \sum_s \frac{P(\pi|x,s;\boldsymbol{\theta}^\pi)q(s|x,\pi^*,ev)}{P(\pi|x,\pi^*,ev)}P(d_j|x,\pi,s) \quad \because \text{Assumption}$$

Back

Derivation: Transition Prob for OUD

$$
\begin{aligned}
&P_s(s'|x, \pi, \pi^*, ev, d_j) \\
&= \sum_s q(s|x, \pi, \pi^*, ev, d_j) f_s(s'|s, rx, d_o^{rx}, d_o^{il}; \boldsymbol{\theta}^s) \\
&= \sum_s \frac{P(d_j|s, x, \pi, \pi^*, ev)}{P(d_j|x, \pi, \pi^*, ev)} q(s|x, \pi, \pi^*, ev) f_s(s'|s, rx, d_o^{rx}, d_o^{il}; \boldsymbol{\theta}^s) \\
&= \sum_s \frac{P(d_j|s, x, \pi)}{P(d_j|x, \pi, \pi^*, ev)} \frac{P(\pi|x, s; \boldsymbol{\theta}^\pi) q(s|x, \pi^*, ev)}{P(\pi|x, \pi^*, ev)} f_s(s'|s, rx, d_o^{rx}, d_o^{il}; \boldsymbol{\theta}^s)
\end{aligned}
$$

Back

Identification: Unrestricted CCPs

$$P(d_j|x,\pi,\pi^*,ev) = \sum_s \frac{P(\pi|x,s;\boldsymbol{\theta}^\pi)q(s|x,\pi^*,ev)}{P(\pi|x,\pi^*,ev)} P(d_j|x,\pi,s)$$

- Let $iv = (\pi^*, ev)$ and suppose $iv \in \{0,1\}$. For a given (d_j, x, π, iv) and $q(s|x,iv)$:

$$\underbrace{\begin{bmatrix} q(s_1|x,iv_0) & 1-q(s_1|x,iv_0) \\ q(s_1|x,iv_1) & 1-q(s_1|x,iv_1) \end{bmatrix}}_{:=Q} \underbrace{\begin{bmatrix} P(d_j|x,s_1) \\ P(d_j|x,s_0) \end{bmatrix}}_{:=p_s} = \underbrace{\begin{bmatrix} P(d_j|x,iv_0) \\ P(d_j|x,iv_1) \end{bmatrix}}_{:=p_x}$$

- If $q(s_1|x,iv_0) \neq q(s_1|x,iv_1)$, then $det(Q) \neq 0$. Thus, $p_s = Q^{-1}p_x$.
- With more variation in iv, $q(s|x,iv)$ is over-identified.

Back

Probability of Dying

- The probability of dying increases wrt (d_o^{rx}, d_o^{il}):

$$f_d(s', a, d_{rx}, d_{il}; \theta^d) = \sum_{i \in D} f(dth_i | s', a, g, d_{rx}, d_{il}; \boldsymbol{\theta}_i^d)$$

where $D = \{ocd, rx, il, bth\}$

$$\begin{aligned}
f(dth_{ocd} | s', a, g, d_{rx}, d_{il}; \theta^{ocd}) &= \frac{\exp(x'_{ocd}\theta^d)}{1 + \exp(x'_{ocd}\theta^d)} \\
f(dth_{rx} | s', a, g, d_{rx}, d_{il}; \theta^{rx}) &= \theta^d_{rx} d_{rx} \\
f(dth_{il} | s', a, g, d_{rx}, d_{il}; \theta^{il}) &= \theta^d_{il} d_{il} \\
f(dth_{bth} | s', a, g, d_{rx}, d_{il}; \theta^{bth}) &= \theta^d_{bth} d_{rx} d_{il} \\
f(dth | s') &= \theta^d_s s'
\end{aligned}$$

Transition

Id: Transition Probabilities

NVSS, NSDUH, SIPP, MEPS

$$P_d^{ocd}(a,g) = \sum_{s'} f_d(a, d_o^{rx}, d_o^{il}; \boldsymbol{\theta}_d) P(s', d_o^{rx} = 0, d_o^{il} = 0 | a, g)$$

$$P_d^{OD:rx} = \sum_{s', d_o^{il}} f_d(a, d_o^{rx}, d_o^{il}; \boldsymbol{\theta}_d) P(s', d_o^{rx} = 1, d_o^{il})$$

$$P_d^{OD:il} = \sum_{s', d_o^{rx}} f_d(a, d_o^{rx}, d_o^{il}; \boldsymbol{\theta}_d) P(s', d_o^{rx}, d_o^{il} = 1)$$

$$P_d^{OD:bth} = \sum_{s'} f_d(a, d_o^{rx}, d_o^{il}; \boldsymbol{\theta}_d) P(s', d_o^{rx} = 1, d_o^{il} = 1)$$

Back

Id: Transition Probabilities

NVSS, NSDUH, SIPP, MEPS

$$P(cw', bh' | cw, bh, d_w, g) = \sum_{d_{opi}, rx, rx', s'} \begin{bmatrix} P(d_{opi}, rx, s' | cw, bh, d_w, g) \tilde{f}_d(a, d_j; \boldsymbol{\theta}_d) \\ f(cw', bh' | cw, bh, rx, d_j, s') \\ f(rx' | cw', bh', rx, d_j, s') \end{bmatrix}$$

$$P(cw', bh', rx' | cw, bh, rx, d_w) = \sum_{d_{opi}, s'} \begin{bmatrix} P(d_{opi}, s' | cw, bh, rx, d_w) \tilde{f}_d(a, d_j; \boldsymbol{\theta}_d) \\ f(cw', bh' | cw, bh, rx, d_j, s') \\ f(rx' | cw', bh', rx, d_j, s') \end{bmatrix}$$

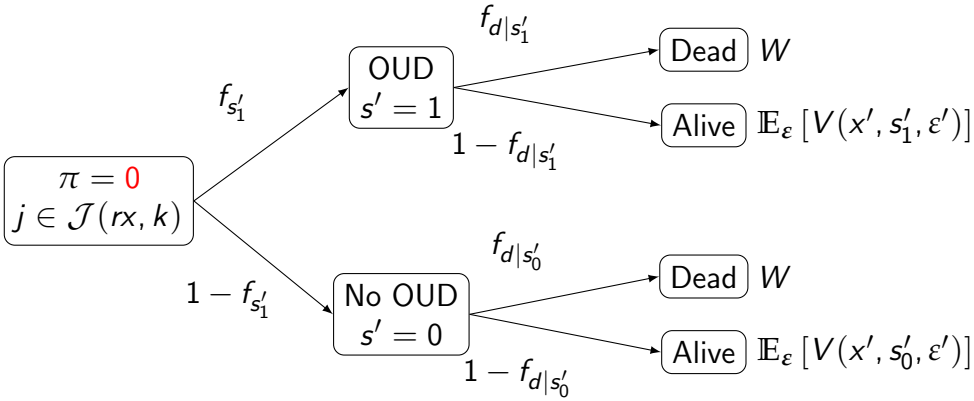
$$P(cw', rx', bh') = \sum_{cw, rx, bh, d_j, s'} \begin{bmatrix} P(cw, rx, bh, d_j, s') \tilde{f}_d(a, d_j; \boldsymbol{\theta}_d) \\ f(cw', bh' | cw, bh, rx, d_j, s') \\ f(rx' | cw', bh', rx, d_j, s') \end{bmatrix}$$

Back

Dynamic Model of Work and Opioid Use

Perception of the risk of "misusing" opioids

- In each period, the individual perceives the risk of "misusing" opioids, π.
  - High risk ($\pi = 0$): expects the actual probability of getting OUD
  - Low risk ($\pi = 1$): expects a lower probability of getting OUD



Simple

Dynamic Model of Work and Opioid Use

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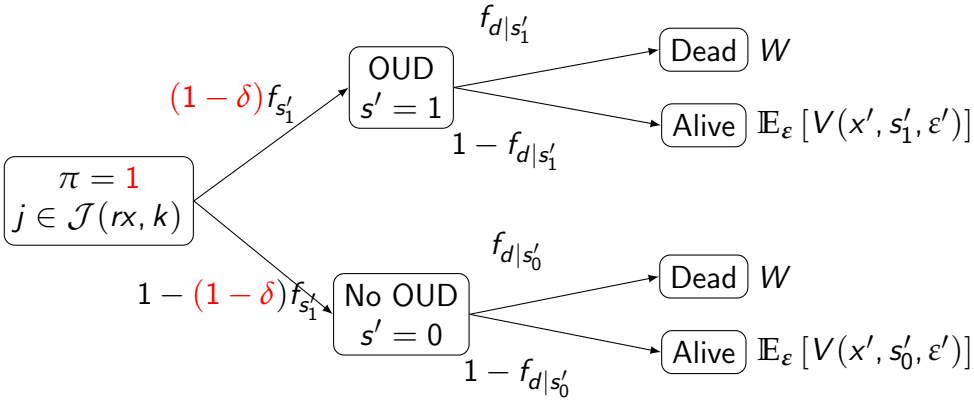

Simple

CVF Differences

- Hotz-Miller inversion implies:

$$\log \frac{P(d_j|x,s)}{P(d_1|x,s)} = v_j(x,s) - v_1(x,s)$$
$$= (\bar{y}_j(\boldsymbol{\theta}^y) + u_{o,j}(\boldsymbol{\theta}^o)) - (\bar{y}_1(\boldsymbol{\theta}^y) - u_{o,1}(\boldsymbol{\theta}^o))$$
$$+ \beta \begin{bmatrix} (1-\pi\delta) f_{s_1',j} \left(f_{d|s_1',j} W + \sum_{x'} \bar{V}(x', s_1') f_{x'|x,s_1',j}(\boldsymbol{\theta}^{ch}, \boldsymbol{\theta}^{rx}) \right) \\ +[1-(1-\pi\delta) f_{s_1',j})] \left(f_{d|s_0',j} W + \sum_{x'} \bar{V}(x', s_0') f_{x'|x,s_0',j}(\boldsymbol{\theta}^{ch}, \boldsymbol{\theta}^{rx}) \right) \end{bmatrix}$$
$$- \beta \begin{bmatrix} (1-\pi\delta) f_{s_1',1} \left(f_{d|s_1',1} W + \sum_{x'} \bar{V}(x', s_1') f_{x'|x,s_1',1}(\boldsymbol{\theta}^{ch}, \boldsymbol{\theta}^{rx}) \right) \\ +[1-(1-\pi\delta) f_{s_1',1})] \left(f_{d|s_0',1} W + \sum_{x'} \bar{V}(x', s_0') f_{x'|x,s_0',1}(\boldsymbol{\theta}^{ch}, \boldsymbol{\theta}^{rx}) \right) \end{bmatrix}$$

Back

Derivation

- Rearranging the RHS:

$$\log \frac{P(d_j|x,s)}{P(d_1|x,s)} = v_j(x,s) - v_1(x,s)$$

$$= (\bar{y}_j(\boldsymbol{\theta}^y) + u_{o,j}(\boldsymbol{\theta}^o)) - (\bar{y}_1(\boldsymbol{\theta}^y) - u_{o,1}(\boldsymbol{\theta}^o)) \qquad \text{Today's Utility}$$

$$+ \beta W \begin{bmatrix} f_{d|s_0',j} + (1-\pi\delta) f_{s_1'|j} \left(f_{d|s_1',j} - f_{d|s_0',j} \right) \\ -f_{d|s_0',1} - (1-\pi\delta) f_{s_1'|1} \left(f_{d|s_1',1} - f_{d|s_0',1} \right) \end{bmatrix} \qquad \text{Death}$$

$$+ \beta(1-\pi\delta) \sum_{x'} \bar{V}(x', s_1') \begin{bmatrix} f_{x'|x,s_1',j}(1 - f_{d|s_1',j}) f_{s_1',j} \\ -f_{x'|x,s_1',1}(1 - f_{d|s_1',1}) f_{s_1',1} \end{bmatrix} \qquad \text{Tomorrow's Life with OUD}$$

$$+ \beta \sum_{x'} \bar{V}(x', s_0') \begin{bmatrix} f_{x'|x,s_0',j}(1 - f_{d|s_0',j}) \left(1 - (1-\pi\delta) f_{s_1',j} \right) \\ f_{x'|x,s_0',1}(1 - f_{d|s_0',1}) \left(1 - (1-\pi\delta) f_{s_1',1} \right) \end{bmatrix} \qquad \text{Tomorrow's Life without OUD}$$

Value Function: Alternative

$$V(x, s, \pi, \varepsilon) = \max_{j \in \mathcal{J}(cw, rx)} u_j(x, s; \boldsymbol{\theta}^y, \boldsymbol{\theta}^o) + \varepsilon_j$$ Today's Utility

$$+ \beta W f_{d|s_1', j}(1 - \pi\delta) f_{s_1'|j}$$ Death with OUD

$$+ \beta \sum_{x', \pi'} \bar{V}(x', s_1', \pi') \left[f_{x'|x, s_1', j}(1 - f_{d|s_1', j})(1 - \pi\delta) f_{s_1', j} \right]$$ Tomorrow's Life with OUD

$$+ \beta W f_{d|s_0', j}(1 - (1 - \pi\delta) f_{s_1'|j})$$ Death without OUD

$$+ \beta \sum_{x', \pi'} \bar{V}(x', s_0', \pi') \left[f_{x'|x, s_0', j}(1 - f_{d|s_0', j}) \left(1 - (1 - \pi\delta) f_{s_1', j}\right) \right]$$ Tomorrow's Life without OUD

Back

Derivation: CVF with future CVF weights

$$v_j(x,s,\pi) = u_j(x,s,\theta^o)$$ Flow utility in $t+1$

$$+\beta\left[W(1-\pi\delta)f_{s_1'|s,j}f_{d'|x,s_1',j} + W(1-(1-\pi\delta)f_{s_1'|s,j})f_{d'|x,s_0',j}\right]$$ Death in $t+1$

$$+\beta\sum_{x',\pi'}\sum_{k\in\mathcal{J}(x')}\begin{bmatrix} u_k(x',s_1',\pi')+ \\ \psi_k(\mathbf{p}(x',s_1',\pi'))\end{bmatrix}\omega_{k,s_1'}f_{x',\pi'|x,s_1',j}(1-\pi\delta)f_{s_1'|x,s,j}\tilde{f}_{d'|x,s',j}$$ Flow utility w/ OUD in $t+1$

$$+\beta\sum_{x',\pi'}\sum_{k\in\mathcal{J}(x')}\begin{bmatrix} u_k(x',s_0',\pi')+ \\ \psi_k(\mathbf{p}(x',s_0',\pi'))\end{bmatrix}\omega_{k,s_0'}f_{x',\pi'|x,s_0',j}(1-(1-\pi\delta)f_{s_1'|x,s,j})\tilde{f}_{d'|x,s',j}$$ Flow utility w/o OUD in $t+1$

$$+\beta^2\sum_{x',\pi'}\sum_{k\in\mathcal{J}(x')}\begin{bmatrix} W(1-\pi'\delta)f_{s_1''|s_1',k}f_{d''|x',s_1'',k}+ \\ W(1-(1-\pi'\delta)f_{s_1''|s_1',k})f_{d''|x',s_0'',k}\end{bmatrix}\omega_{k,s_1'}f_{x',\pi'|x,s_1',j}(1-\pi\delta)f_{s_1'|x,s,j}\tilde{f}_{d'|x,s',j}$$ Death w/ OUD in $t+1$

$$+\beta^2\sum_{x',\pi'}\sum_{k\in\mathcal{J}(x')}\begin{bmatrix} W(1-\pi'\delta)f_{s_1''|s_0',k}f_{d''|x',s_1'',k}+ \\ W(1-(1-\pi'\delta)f_{s_1''|s_0',k})f_{d''|x',s_0'',k}\end{bmatrix}\omega_{k,s_0'}f_{x',\pi'|x,s_1',j}(1-(1-\pi\delta)f_{s_1'|x,s,j})\tilde{f}_{d'|x,s',j}$$ Death w/o OUD in $t+2$

$$+\beta^2\sum_{x'',s''}\sum_{x'}\sum_{k\in\mathcal{J}(x')}\begin{bmatrix} V(x'',s_1'',\pi'')f_{x''|x',s_1',k}f_{d''|s_1''}(1-\pi')f_{s_1''|s_1'}+ \\ V(x'',s_0'',\pi'')f_{x''|x',s_0',k}f_{d''|s_0''}(1-(1-\pi'))f_{s_1''|s_1'}\end{bmatrix}\omega_{k,s_0'}f_{x|s_1'}(1-\pi\delta)f_{s_1'|s,j}\tilde{f}_{d|s',j}$$ Survival w/ OUD in $t+2$

$$+\beta^2\sum_{x'',s''}\sum_{x'}\sum_{k\in\mathcal{J}(x')}\begin{bmatrix} V(x'',s_1'',\pi'')f_{x''|x',s_1',k}f_{d''|s_1''}(1-\pi')f_{s_1''|s_0'}+ \\ V(x'',s_0'',\pi'')f_{x''|x',s_0',k}f_{d''|s_0''}(1-(1-\pi'))f_{s_1''|s_0'}\end{bmatrix}\omega_{k,s_1'}f_{x|s_0'}(1-(1-\pi\delta)f_{s_1'|s,j})\tilde{f}_{d|s',j}$$ Survival w/o OUD $t+2$

where $\omega_{k,s_1'} = \omega_k(x',s_1',\pi',x,s,\pi,j)$ and $\omega_{k,s_0'} = \omega_k(x',s_0',\pi',x,s,\pi,j)$

Back

CVF Differences with Finite Dependence

If a pair of choices $(1, j)$ at (x, s, π) exhibit a 1-finite dependence, then:

$$\psi_1(\mathbf{p}(x,s,\pi)) - \psi_j(\mathbf{p}(x,s,\pi)) = v_j(x,s,\pi) - v_1(x,s,\pi)$$

$$= u_j(x,s;\boldsymbol{\theta}^y,\theta^o) - u_1(x,s;\boldsymbol{\theta}^y,\theta^o) \qquad \Delta \text{ Flow utility in } t$$

$$+\beta\left[W(1-\pi\delta)f_{s_1'|s,j}f_{d'|x,s_1',j} + W(1-(1-\pi\delta)f_{s_1'|s,j})f_{d'|x,s_0',j}\right] \qquad \text{Death in } t+1 \text{ given } j$$

$$-\beta\left[W(1-\pi\delta)f_{s_1'|s,1}f_{d'|x,s_1',j} + W(1-(1-\pi\delta)f_{s_1'|s,1})f_{d'|x,s_0',1}\right] \qquad \text{Death in } t+1 \text{ given } 1$$

$$+\beta\sum_{x',\pi'}\sum_{k\in\mathcal{J}(x')}\begin{bmatrix}u_k(x',s_1';\boldsymbol{\theta}^y,\theta^o)+\\ \psi_k(\mathbf{p}(x',s_1',\pi'))\end{bmatrix}\omega_{k,s_1',j}f_{x',\pi'|x,s_1',j}(1-\pi\delta)f_{s_1'|x,s,j}\tilde{f}_{d'|x,s',j} \qquad \text{Flow utility w/ OUD in } t+1 \text{ given } j$$

$$-\beta\sum_{x',\pi'}\sum_{k\in\mathcal{J}(x')}\begin{bmatrix}u_k(x',s_1';\boldsymbol{\theta}^y,\theta^o)+\\ \psi_k(\mathbf{p}(x',s_1',\pi'))\end{bmatrix}\omega_{k,s_1',1}f_{x',\pi'|x,s_1',j}(1-\pi\delta)f_{s_1'|x,s,1}\tilde{f}_{d'|x,s',1} \qquad \text{Flow utility w/ OUD in } t+1 \text{ given } 1$$

$$+\beta\sum_{x',\pi'}\sum_{k\in\mathcal{J}(x')}\begin{bmatrix}u_k(x',s_0';\boldsymbol{\theta}^y,\theta^o)+\\ \psi_k(\mathbf{p}(x',s_0',\pi'))\end{bmatrix}\omega_{k,s_0',j}f_{x',\pi'|x,s_0',j}(1-(1-\pi\delta)f_{s_1'|x,s,j})\tilde{f}_{d'|x,s',j} \qquad \text{Flow utility w/o OUD in } t+1 \text{ given } j$$

$$-\beta\sum_{x',\pi'}\sum_{k\in\mathcal{J}(x')}\begin{bmatrix}u_k(x',s_0';\boldsymbol{\theta}^y,\theta^o)+\\ \psi_k(\mathbf{p}(x',s_0',\pi'))\end{bmatrix}\omega_{k,s_0',1}f_{x',\pi'|x,s_0',1}(1-(1-\pi\delta)f_{s_1'|x,s,1})\tilde{f}_{d'|x,s',1} \qquad \text{Flow utility w/ OUD in } t+1 \text{ given } 1$$

where $\omega_{k,s',j} = \omega_k(x',s',\pi',x,s,\pi,j;\delta)$

Back

Estimation: Stage 1-1

- The likelihood of observing $(x_n, \pi_n, d_{j,n}, y_{k,n}, s'_n, ev_n, \pi^*_n)$ is

$$l(x_n, \pi_n, d_{j,n}, y_n, s'_n; \boldsymbol{\theta}) =$$

$$\sum_{s=0,1} q(s|x_n, ev, \pi^*; \boldsymbol{\theta}^u) \prod_{j,\pi,y_k,s'} \begin{bmatrix} f_\pi(\pi|x_\pi, s; \boldsymbol{\theta}^\pi) \times \\ P(d_{j,n}|x_n, \pi_n, s) \times \\ F_y(y_{k,n}|d_{j,n}, x_n, s; \boldsymbol{\theta}^y, \sigma_y) \times \\ f_s(s'|d_{j,n}, rx_n, s; \boldsymbol{\theta}^s) \end{bmatrix}^{\mathbf{1}\{\pi_n, d_{jn}, y_{kn}, s'_n\}}$$

where $F_y(y_k|d_{j,n}, x_n, s; \boldsymbol{\theta}^y) = \Phi\left(\frac{y_{k+1,n} - \bar{y}(x_y, d_j; \boldsymbol{\theta}^y)}{\sigma_y}\right) - \Phi\left(\frac{y_{k,n} - \bar{y}(x_y, d_j; \boldsymbol{\theta}^y)}{\sigma_y}\right)$

Back

Estimation: Stage 1-2

- Given $\hat{\boldsymbol{\theta}}^u$, $\hat{P}(d_j|x,\pi,s)$, $\hat{\boldsymbol{\theta}}^\pi$, $\hat{\boldsymbol{\theta}}^s$, estimate $\boldsymbol{\theta}^d$, $\boldsymbol{\theta}^{ch}$, $\boldsymbol{\theta}^{rx}$, γ, $\boldsymbol{\alpha}$
- NVSS, NSDUH, SIPP, MEPS

$$f_d(a,g) = \sum_{s',d_o^{rx},d_o^{il}} f_d(\theta_d) P(s', d_o^{rx}, d_o^{il}|a,g)$$

$$P(cw', bh'|cw, bh, d_w, g) = \sum_{d_{opi},rx,rx',s'} \begin{bmatrix} P(d_{opi}, rx, s'|cw, bh, d_w, g)\tilde{f}_d(a, d_j; \theta_d) \\ f(cw', bh', rx'|cw, bh, rx, d_j, s') \end{bmatrix}$$

$$P(cw', bh', rx'|cw, bh, rx, d_w) = \sum_{d_{opi},s'} \begin{bmatrix} P(d_{opi}, s'|cw, bh, rx, d_w)\tilde{f}_d(a, d_j; \theta_d) \\ f(cw', bh'|cw, bh, rx, s', d_j; \boldsymbol{\theta}^{ch}) \\ f(rx'|cw', bh', rx, s', d_j; \boldsymbol{\theta}^{rx}, \gamma, \boldsymbol{\alpha}) \end{bmatrix}$$

$$P(cw', rx', bh') = \sum_{cw,rx,bh,d_j,s'} \begin{bmatrix} P(cw, rx, bh, d_j, s')\tilde{f}_d(a, d_j; \theta_d) \\ f(cw', bh'|cw, bh, rx, s', d_j; \boldsymbol{\theta}^{ch}) \\ f(rx'|cw', bh', rx, s', d_j; \boldsymbol{\theta}^{rx}, \gamma, \boldsymbol{\alpha}) \end{bmatrix}$$

Back Back

Estimation: Stage 1-2

- Given $\hat{\boldsymbol{\theta}}^u$, $\hat{P}(d_j|x,\pi,s)$, $\hat{\boldsymbol{\theta}}^\pi$, $\hat{\boldsymbol{\theta}}^s$, estimate $\boldsymbol{\theta}^d$, $\boldsymbol{\theta}^{ch}$, $\boldsymbol{\theta}^{rx}$, γ, $\boldsymbol{\alpha}$
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$$f_d(a,g) = \sum_{s',d_o^{rx},d_o^{il}} f_d(\boldsymbol{\theta}_d) P(s', d_o^{rx}, d_o^{il}|a,g)$$

$$P(cw', bh'|cw, bh, d_w, g) = \sum_{d_{opi},rx,rx',s'} \begin{bmatrix} P(d_{opi}, rx, s'|cw, bh, d_w, g)\tilde{f}_d(a, d_j; \theta_d) \\ f(cw', bh', rx'|cw, bh, rx, d_j, s') \end{bmatrix}$$

$$P(cw', bh', rx'|cw, bh, rx, d_w) = \sum_{d_{opi},s'} \begin{bmatrix} P(d_{opi}, s'|cw, bh, rx, d_w)\tilde{f}_d(a, d_j; \theta_d) \\ f(cw', bh'|cw, bh, rx, s', d_j; \boldsymbol{\theta}^{ch}) \\ f(rx'|cw', bh', rx, s', d_j; \boldsymbol{\theta}^{rx}, \gamma, \boldsymbol{\alpha}) \end{bmatrix}$$

$$P(cw', rx', bh') = \sum_{cw,rx,bh,d_j,s'} \begin{bmatrix} P(cw, rx, bh, d_j, s')\tilde{f}_d(a, d_j; \theta_d) \\ f(cw', bh'|cw, bh, rx, s', d_j; \boldsymbol{\theta}^{ch}) \\ f(rx'|cw', bh', rx, s', d_j; \boldsymbol{\theta}^{rx}, \gamma, \boldsymbol{\alpha}) \end{bmatrix}$$

Back Back

Estimation: Stage 1-2

- Given $\hat{\boldsymbol{\theta}}^u$, $\hat{P}(d_j|x,\pi,s)$, $\hat{\boldsymbol{\theta}}^\pi$, $\hat{\boldsymbol{\theta}}^s$, estimate $\boldsymbol{\theta}^d$, $\boldsymbol{\theta}^{ch}$, $\boldsymbol{\theta}^{rx}$, γ, $\boldsymbol{\alpha}$
- NVSS, NSDUH, SIPP, MEPS

$$f_d(a,g) = \sum_{s',d_o^{rx},d_o^{il}} f_d(\boldsymbol{\theta}_d) P(s', d_o^{rx}, d_o^{il}|a,g)$$

$$P(cw', bh'|cw, bh, d_w, g) = \sum_{d_{opi},rx,rx',s'} \begin{bmatrix} P(d_{opi}, rx, s'|cw, bh, d_w, g)\tilde{f}_d(a, d_j; \theta_d) \\ f(cw', bh', rx'|cw, bh, rx, d_j, s') \end{bmatrix}$$

$$P(cw', bh', rx'|cw, bh, rx, d_w) = \sum_{d_{opi},s'} \begin{bmatrix} P(d_{opi}, s'|cw, bh, rx, d_w)\tilde{f}_d(a, d_j; \theta_d) \\ f(cw', bh'|cw, bh, rx, s', d_j; \boldsymbol{\theta}^{ch}) \\ f(rx'|cw', bh', rx, s', d_j; \boldsymbol{\theta}^{rx}, \gamma, \boldsymbol{\alpha}) \end{bmatrix}$$

$$P(cw', rx', bh') = \sum_{cw,rx,bh,d_j,s'} \begin{bmatrix} P(cw, rx, bh, d_j, s')\tilde{f}_d(a, d_j; \theta_d) \\ f(cw', bh'|cw, bh, rx, s', d_j; \boldsymbol{\theta}^{ch}) \\ f(rx'|cw', bh', rx, s', d_j; \boldsymbol{\theta}^{rx}, \gamma, \boldsymbol{\alpha}) \end{bmatrix}$$

Back Back

Estimation: Stage 1-2

- Given $\hat{\boldsymbol{\theta}}^u$, $\hat{P}(d_j|x,\pi,s)$, $\hat{\boldsymbol{\theta}}^\pi$, $\hat{\boldsymbol{\theta}}^s$, estimate $\boldsymbol{\theta}^d$, $\boldsymbol{\theta}^{ch}$, $\boldsymbol{\theta}^{rx}$, γ, $\boldsymbol{\alpha}$
- NVSS, NSDUH, SIPP, MEPS

$$f_d(a,g) = \sum_{s',d_o^{rx},d_o^{il}} f_d(\boldsymbol{\theta}_d) P(s', d_o^{rx}, d_o^{il}|a,g)$$

$$P(cw', bh'|cw, bh, d_w, g) = \sum_{d_{opi},rx,rx',s'} \begin{bmatrix} P(d_{opi}, rx, s'|cw, bh, d_w, g)\tilde{f}_d(a, d_j; \theta_d) \\ f(cw', bh', rx'|cw, bh, rx, d_j, s') \end{bmatrix}$$

$$P(cw', bh', rx'|cw, bh, rx, d_w) = \sum_{d_{opi},s'} \begin{bmatrix} P(d_{opi}, s'|cw, bh, rx, d_w)\tilde{f}_d(a, d_j; \theta_d) \\ f(cw', bh'|cw, bh, rx, s', d_j; \boldsymbol{\theta}^{ch}) \\ f(rx'|cw', bh', rx, s', d_j; \boldsymbol{\theta}^{rx}, \gamma, \boldsymbol{\alpha}) \end{bmatrix}$$

$$P(cw', rx', bh') = \sum_{cw,rx,bh,d_j,s'} \begin{bmatrix} P(cw, rx, bh, d_j, s')\tilde{f}_d(a, d_j; \theta_d) \\ f(cw', bh'|cw, bh, rx, s', d_j; \boldsymbol{\theta}^{ch}) \\ f(rx'|cw', bh', rx, s', d_j; \boldsymbol{\theta}^{rx}, \gamma, \boldsymbol{\alpha}) \end{bmatrix}$$

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Estimation: Stage 1-2

- Given $\hat{\boldsymbol{\theta}}^u$, $\hat{P}(d_j|x,\pi,s)$, $\hat{\boldsymbol{\theta}}^\pi$, $\hat{\boldsymbol{\theta}}^s$, estimate $\boldsymbol{\theta}^d$, $\boldsymbol{\theta}^{ch}$, $\boldsymbol{\theta}^{rx}$, γ, $\boldsymbol{\alpha}$
- NVSS, NSDUH, SIPP, MEPS

$$f_d(a,g) = \sum_{s',d_o^{rx},d_o^{il}} f_d(\boldsymbol{\theta}_d) P(s', d_o^{rx}, d_o^{il}|a,g)$$

$$P(cw', bh'|cw, bh, d_w, g) = \sum_{d_{opi},rx,rx',s'} \begin{bmatrix} P(d_{opi}, rx, s'|cw, bh, d_w, g)\tilde{f}_d(a, d_j; \theta_d) \\ f(cw', bh', rx'|cw, bh, rx, d_j, s') \end{bmatrix}$$

$$P(cw', bh', rx'|cw, bh, rx, d_w) = \sum_{d_{opi},s'} \begin{bmatrix} P(d_{opi}, s'|cw, bh, rx, d_w)\tilde{f}_d(a, d_j; \theta_d) \\ f(cw', bh'|cw, bh, rx, s', d_j; \boldsymbol{\theta}^{ch}) \\ f(rx'|cw', bh', rx, s', d_j; \boldsymbol{\theta}^{rx}, \gamma, \boldsymbol{\alpha}) \end{bmatrix}$$

$$P(cw', rx', bh') = \sum_{cw,rx,bh,d_j,s'} \begin{bmatrix} P(cw, rx, bh, d_j, s')\tilde{f}_d(a, d_j; \theta_d) \\ f(cw', bh'|cw, bh, rx, s', d_j; \boldsymbol{\theta}^{ch}) \\ f(rx'|cw', bh', rx, s', d_j; \boldsymbol{\theta}^{rx}, \gamma, \boldsymbol{\alpha}) \end{bmatrix}$$

Back Back

Estimation: Stage 2

- Using the one-to-one mapping between CCPs and CVF differences:

$$\psi_1(x,s) - \psi_j(x,s) = v_j(x,s) - v_1(x,s)$$

$$= (\bar{y}_j(\boldsymbol{\theta}^y) + u_{o,j}(\boldsymbol{\theta}^o)) - (\bar{y}_1(\boldsymbol{\theta}^y) - u_{o,1}(\boldsymbol{\theta}^o))$$ Today's Utility

$$+ \beta W \begin{bmatrix} f_{d|s_0',j} + (1-\pi\delta) f_{s_1'|j} \left(f_{d|s_1',j} - f_{d|s_0',j} \right) \\ -f_{d|s_0',1} - (1-\pi\delta) f_{s_1'|1} \left(f_{d|s_1',1} - f_{d|s_0',1} \right) \end{bmatrix}$$ Death

$$+ \beta(1-\pi\delta) \sum_{x'} \bar{V}(x', s_1') \begin{bmatrix} f_{x'|x,s_1',j}(1 - f_{d|s_1',j}) f_{s_1',j} \\ -f_{x'|x,s_1',1}(1 - f_{d|s_1',1}) f_{s_1',1} \end{bmatrix}$$ Tomorrow's Life with OUD

$$+ \beta \sum_{x'} \bar{V}(x', s_0') \begin{bmatrix} f_{x'|x,s_0',j}(1 - f_{d|s_0',j}) \left(1 - (1-\pi\delta) f_{s_1',j} \right) \\ f_{x'|x,s_0',1}(1 - f_{d|s_0',1}) \left(1 - (1-\pi\delta) f_{s_1',1} \right) \end{bmatrix}$$ Tomorrow's Life without OUD

Identification Derivation Back

Estimation: Stage 2

- Using the one-to-one mapping between CCPs and CVF differences:

$$\psi_1(x,s) - \psi_j(x,s) = v_j(x,s) - v_1(x,s)$$

$$= (\bar{y}_j(\boldsymbol{\theta}^y) + u_{o,j}(\boldsymbol{\theta}^o)) - (\bar{y}_1(\boldsymbol{\theta}^y) - u_{o,1}(\boldsymbol{\theta}^o))$$ Today's Utility

$$+ \beta W \begin{bmatrix} f_{d|s_0',j} + (1-\pi\delta) f_{s_1'|j} \left(f_{d|s_1',j} - f_{d|s_0',j} \right) \\ -f_{d|s_0',1} - (1-\pi\delta) f_{s_1'|1} \left(f_{d|s_1',1} - f_{d|s_0',1} \right) \end{bmatrix}$$ Death

$$+ \beta(1-\pi\delta) \sum_{x'} \bar{V}(x', s_1') \begin{bmatrix} f_{x'|x,s_1',j}(1 - f_{d|s_1',j}) f_{s_1',j} \\ -f_{x'|x,s_1',1}(1 - f_{d|s_1',1}) f_{s_1',1} \end{bmatrix}$$ Tomorrow's Life with OUD

$$+ \beta \sum_{x'} \bar{V}(x', s_0') \begin{bmatrix} f_{x'|x,s_0',j}(1 - f_{d|s_0',j}) \left(1 - (1-\pi\delta) f_{s_1',j} \right) \\ f_{x'|x,s_0',1}(1 - f_{d|s_0',1}) \left(1 - (1-\pi\delta) f_{s_1',1} \right) \end{bmatrix}$$ Tomorrow's Life without OUD

Identification Derivation Back

Estimation: Stage 2

- Using the one-to-one mapping between CCPs and CVF differences:

$$\psi_1(x,s) - \psi_j(x,s) = v_j(x,s) - v_1(x,s)$$

$$= (\bar{y}_j(\boldsymbol{\theta}^y) + u_{o,j}(\boldsymbol{\theta}^o)) - (\bar{y}_1(\boldsymbol{\theta}^y) - u_{o,1}(\boldsymbol{\theta}^o))$$ Today's Utility

$$+ \beta W \begin{bmatrix} f_{d|s_0',j} + (1-\pi\delta) f_{s_1'|j} \left(f_{d|s_1',j} - f_{d|s_0',j} \right) \\ -f_{d|s_0',1} - (1-\pi\delta) f_{s_1'|1} \left(f_{d|s_1',1} - f_{d|s_0',1} \right) \end{bmatrix}$$ Death

$$+ \beta(1-\pi\delta) \sum_{x'} \bar{V}(x', s_1') \begin{bmatrix} f_{x'|x,s_1',j}(1 - f_{d|s_1',j}) f_{s_1',j} \\ -f_{x'|x,s_1',1}(1 - f_{d|s_1',1}) f_{s_1',1} \end{bmatrix}$$ Tomorrow's Life with OUD

$$+ \beta \sum_{x'} \bar{V}(x', s_0') \begin{bmatrix} f_{x'|x,s_0',j}(1 - f_{d|s_0',j}) \left(1 - (1-\pi\delta) f_{s_1',j} \right) \\ f_{x'|x,s_0',1}(1 - f_{d|s_0',1}) \left(1 - (1-\pi\delta) f_{s_1',1} \right) \end{bmatrix}$$ Tomorrow's Life without OUD

Identification Derivation Back

Estimation: Stage 2

- Using the one-to-one mapping between CCPs and CVF differences:

$$\psi_1(x,s) - \psi_j(x,s) = v_j(x,s) - v_1(x,s)$$

$$= (\bar{y}_j(\boldsymbol{\theta}^y) + u_{o,j}(\boldsymbol{\theta}^o)) - (\bar{y}_1(\boldsymbol{\theta}^y) - u_{o,1}(\boldsymbol{\theta}^o))$$ Today's Utility

$$+ \beta W \begin{bmatrix} f_{d|s_0',j} + (1-\pi\delta) f_{s_1'|j} \left(f_{d|s_1',j} - f_{d|s_0',j} \right) \\ -f_{d|s_0',1} - (1-\pi\delta) f_{s_1'|1} \left(f_{d|s_1',1} - f_{d|s_0',1} \right) \end{bmatrix}$$ Death

$$+ \beta(1-\pi\delta) \sum_{x'} \bar{V}(x', s_1') \begin{bmatrix} f_{x'|x,s_1',j}(1 - f_{d|s_1',j}) f_{s_1',j} \\ -f_{x'|x,s_1',1}(1 - f_{d|s_1',1}) f_{s_1',1} \end{bmatrix}$$ Tomorrow's Life with OUD

$$+ \beta \sum_{x'} \bar{V}(x', s_0') \begin{bmatrix} f_{x'|x,s_0',j}(1 - f_{d|s_0',j}) \left(1 - (1-\pi\delta) f_{s_1',j} \right) \\ f_{x'|x,s_0',1}(1 - f_{d|s_0',1}) \left(1 - (1-\pi\delta) f_{s_1',1} \right) \end{bmatrix}$$ Tomorrow's Life without OUD

Identification Derivation Back

Estimation: Stage 2

- Using the one-to-one mapping between CCPs and CVF differences:

$$
\begin{aligned}
&\psi_1(x,s) - \psi_j(x,s) = v_j(x,s) - v_1(x,s) \\
&= (\bar{y}_j(\boldsymbol{\theta}^y) + u_{o,j}(\boldsymbol{\theta}^o)) - (\bar{y}_1(\boldsymbol{\theta}^y) - u_{o,1}(\boldsymbol{\theta}^o)) && \text{Today's Utility} \\
&+ \beta W \begin{bmatrix} f_{d|s_0',j} + (1-\pi\delta) f_{s_1'|j} \left(f_{d|s_1',j} - f_{d|s_0',j} \right) \\ -f_{d|s_0',1} - (1-\pi\delta) f_{s_1'|1} \left(f_{d|s_1',1} - f_{d|s_0',1} \right) \end{bmatrix} && \text{Death} \\
&+ \beta(1-\pi\delta) \sum_{x'} \bar{V}(x', s_1') \begin{bmatrix} f_{x'|x,s_1',j}(1 - f_{d|s_1',j}) f_{s_1',j} \\ -f_{x'|x,s_1',1}(1 - f_{d|s_1',1}) f_{s_1',1} \end{bmatrix} && \text{Tomorrow's Life with OUD} \\
&+ \beta \sum_{x'} \bar{V}(x', s_0') \begin{bmatrix} f_{x'|x,s_0',j}(1 - f_{d|s_0',j}) \left(1 - (1-\pi\delta) f_{s_1',j} \right) \\ f_{x'|x,s_0',1}(1 - f_{d|s_0',1}) \left(1 - (1-\pi\delta) f_{s_1',1} \right) \end{bmatrix} && \text{Tomorrow's Life without OUD}
\end{aligned}
$$

Identification Derivation Back

Estimation: Stage 2

- For a given $\boldsymbol{\omega}(x', s', x, s)$ where $\sum_{k \in \mathcal{J}(rx', k')} \omega_k(x', s', x, s) = 1$ and $|\omega_k(x', s', x, s)| < \infty$,

$$\bar{V}(x', s_0') = \sum_{k \in \mathcal{J}(rx', k')} \left(v_k(x', s_0') - \log P(d_k | x', s_0')\right) \omega_k(x', s_0', x, s)$$

$$\bar{V}(x', s_1') = \sum_{k \in \mathcal{J}(rx', k')} \left(v_k(x', s_1') - \log P(d_k | x', s_1')\right) \omega_k(x', s_1', x, s)$$

- Iterate $v_k(x', s')$ once more wrt utilities and future value functions
- Find the weights $\omega_k(x'', s_0'', x, s)$ that cancels out the two-period-ahead ex-ante value functions $\bar{V}(x'', s'')$
- Then, we have a moment for identifying $\boldsymbol{\theta}^o$.

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Estimation: Stage 3

- Suppose we have such weights for each $(x, s, j, 1)$.
- Then, estimating $\boldsymbol{\theta}^o, W, \delta$ becomes a GMM problem:

$$\begin{bmatrix} \log P(d_2|x_1, s_0) - \log P(d_1|x_1, s_0) \\ \vdots \\ \log P(d_J|x_1, s_0) - \log P(d_1|x_1, s_0) \\ \log P(d_2|x_1, s_1) - \log P(d_1|x_1, s_1) \\ \vdots \\ \log P(d_J|x_1, s_1) - \log P(d_1|x_1, s_1) \\ \vdots \\ \log P(d_2|x_M, s_1) - \log P(d_1|x_M, s_1) \\ \vdots \\ \log P(d_J|x_M, s_1) - \log P(d_1|x_M, s_1) \end{bmatrix} = \begin{bmatrix} v_2(x_1, s_0) - v_1(x_1, s_0) \\ \vdots \\ v_J(x_1, s_0) - v_1(x_1, s_0) \\ v_2(x_1, s_1) - v_1(x_1, s_1) \\ \vdots \\ v_J(x_1, s_1) - v_1(x_1, s_1) \\ \vdots \\ v_2(x_M, s_1) - v_1(x_M, s_1) \\ \vdots \\ v_J(x_M, s_1) - v_1(x_M, s_1) \end{bmatrix}$$

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